THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 01

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** the same thickness	Keep other factors the same
2	**(b)** thermometer	Measures °C
3	**(a)** fixing atmospheric nitrogen	Increases fertility
4	**(a)** Regular exercise and adequate sleep	Healthy lifestyle
5	**(a)** vaccination and isolation	Viral diseases
6	**(c)** reverse the direction of current	Polarity depends on current direction
7	**(c)** chemicals get used up	Called a dead cell
8	**(c)** positive and negative	Like repel, unlike attract
9	**(b)** 60 N/m²	Pressure = force ÷ area
10	**(b)** increase water pressure at taps	Greater height gives greater pressure
11	**(c)** are free and widely spaced	Large space
12	**(a)** compound (iron oxide)	Iron reacts with moist air
13	**(c)** a naturally occurring element that is a metal	Native mineral
14	**(d)** 1 g/cm³	1000 kg/m³
15	**(c)** reflects sunlight	Not a luminous body
16	**(a)** Thiruvananthapuram	Rocket and launch vehicle technology
17	**(b)** omnivores	e.g. humans, crows
18	**(d)** recycling nutrients	Nutrients return to soil
19	**(c)** biodiversity loss	Three linked threats
20	**(a)** harmful ultraviolet rays	Absorbs UV

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	A fuel made by microbial breakdown of organic waste (1). Methane (1).	1+1
22	Bacteria that were killed earlier survive an antibiotic (1). They multiply despite the treatment (1).	1+1
23	Mobile phones and laptops (1). Inverters or electric vehicles (1).	1+1
24	Stay away from tall objects and crouch low (1). Avoid metal umbrellas and get out of water (1).	1+1
25	Mixtures have variable composition; compounds fixed (1). Mixture components keep properties; compound has new properties (1).	1+1
26	Salt in water (1). Alcohol in water (1).	1+1
27	Clean air and water (1). Food, medicine or climate regulation (1).	1+1
28	Climate change (1). Pollution and biodiversity loss (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	To avoid burns and injury (1). Use adult help (1). Keep distance from splattering hot oil (1).	1+1+1
30	Mass is amount of matter and remains same (1½). Weight depends on gravitational pull of the planet (1½).	1½+1½
31	Solid wax melts into liquid (1). Liquid wax rises through wick and vaporises (1). Particle forces weaken at each stage (1).	1+1+1
32	Rough surface has irregularities (1). Normals point in different directions (1). Reflected rays scatter, giving diffuse reflection (1).	1+1+1
33	Note its position and shape each day (1). Position shifts eastward (1). Shape of the bright part changes (1).	1+1+1
34	Air provides oxygen for respiration (1). Water is needed by all organisms (1). Sunlight supports photosynthesis (1).	1+1+1

35	Rechargeable batteries wear out with many cycles (1½). Their capacity gradually falls (1½).	1½+1½
	OR: Magnetic: current makes a coil act as a magnet (1). Use: crane (½). Heating: wire gets hot (1). Use: heater (½).	1+½+1+½
36	Clean air (1). Fresh water (1). Food and medicine (1).	1+1+1
	OR: They make food using sunlight (1½). All consumers depend directly or indirectly on them (1½).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	Avoids pain (1). Saves cost (1). Prevents spread (1). Vaccines and hygiene (1). Healthy lifestyle reduces NCDs (1).	1+1+1+1+1
	OR: Hygiene prevents contact spread (1). Clean water prevents cholera and typhoid (1½). Sanitation prevents worms like Ascaris (1½). Handwashing and cooked food (1).	1+1½+1½+1
38	Storm surge and floods (1). Heavy rain, landslides (1). Salt ruins soil (1). Early warning and evacuation (1). Strong shelters (1).	1+1+1+1+1
	OR: Press out air (1). Pressure inside decreases (1). Outside pressure is higher (1). Force needed to overcome difference (1). Releasing allows air in (1).	1+1+1+1+1
39	Add solute till no more dissolves (1). Add more solvent (2). Raise temperature (2).	1+2+2
	OR: A = 1 g/mL (1½). B = 0.8 g/mL (1½). C = 1.2 g/mL (1). B floats (1).	1½+1½+1+1
40	Deforestation (1). Pollution (1). Climate change and invasive species (1). Plant trees (1). Reduce, reuse, recycle (1).	1+1+1+1+1
	OR: Vrikshayurveda text (1). Organic manure (1). Kunapa Jala (1). Chemical farming harms soil (1). Balanced approach (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER